P2 — Certified screening of the VV10 nonlocal-correlation double sum: what the bound buys, and what it costs

Angran Xia*
School of Chemistry & Chemical Engineering, Shaanxi Normal University, Xi'an 710119, Shaanxi, China
haangyeon@snnu.edu.cn

Changwei Wang
School of Chemistry & Chemical Engineering, Shaanxi Normal University, Xi'an 710119, Shaanxi, China

Draft — 29 September 2026
ORCIDs PENDING, author-supplied

Authors. Angran Xia (corresponding; haangyeon@snnu.edu.cn) and Changwei Wang — School of Chemistry & Chemical Engineering, Shaanxi Normal University, Chang'an Campus, Xi'an 710119, Shaanxi, China. Names, institution and the corresponding e-mail were supplied by the author on 2026-09-30; the college's English wording is taken from the college's own English site and the campus postal code from the university's contact page. The authors' Chinese names live in paper/software.md's front matter and CITATION.cff, not here: the P1 pipeline typesets with pdflatex, which has no CJK font, and the same rule is applied to all three manuscripts. ORCIDs and the second author's e-mail remain PENDING (author-supplied).

1 Abstract

The VV10 nonlocal-correlation term of ωB97X-V is evaluated as a double sum over real-space integration points, quadratic in the number of points. We implemented a *certified* screening scheme for it: every pair of grid-point blocks is either evaluated exactly or replaced by an upper bound proved to dominate its contribution, the bounds are accumulated into a single pre-declared error, and the grid-point-pair structure is kept intact so that the result can still be attributed fragment by fragment. We then measured the scheme against the dense sum on an unloaded machine, and report that it does not pay. At the accuracy a fragment-pair decomposition needs (1e-6 kcal/mol per pair term), 84.4-95.4% of the far-field point pairs are still evaluated and the screened path takes 1.95-10.25× as long; across a 3-12 atom water-chain ladder no sample is faster than the dense sum, and the wall-clock ratio spans 1.08-10.28×. That window is now backed by a second, independent one on the same box (15 further screened/exact pairs, 5.46-13.29×, again none faster): 33 pairs total across two windows [evidence/scaling/step_j.json, evidence/scaling/step_j_rerun_20260930.json]. The split between the two measurements is worth stating: the *work* side reproduced bit-for-bit in all 15 overlapping system/tolerance pairs (the fraction of far-field point pairs evaluated is a deterministic function of τ), while the wall clock moved by as much as 2.25× between windows on the same inputs (largest: the 3-atom τ=1e-8 run, 0.426 s against 0.957 s) - so the coverage claim is exact and the cost claim is a range, which is why we report it as one. Three measurements explain the

*Corresponding author; e-mail in the byline.

result and are the substance of this paper: at atom-block granularity the loosest block-pair bound in a real dimer exceeds the whole quantity it is supposed to judge by a factor of 2.4e5; the near field is 100% of the cross point-pairs (all root box separations are exactly 0.0000 Å), so a geometric accept/reject rule has a 0.0% saving ceiling; and tightening the bound by splitting each block into a dense core and a sparse tail raises the fraction of droppable point pairs only from 12.349% to 13.650%. A mass-weighted-ball criterion, which looks obviously monotone under refinement, is not — it loses the property on 18% of refinements, which we found only by counting. We state the four conditions a certified scheme would need to meet before it could win, note that none holds for weakly bound dimers, and keep the screening off by default.

2 1. What we built, what it measured, and what was already known

Screening in this paper is an algorithmic sparsity truncation with a proof attached: a block pair is skipped only when an upper bound on its contribution, computed before the skip, dominates everything that would have been lost. It is not the Lindhard/dielectric screening that the vdW-DF literature also calls "screening", and we avoid that reading by never calling the kernel screened — only the sum is.

The VV10 nonlocal-correlation energy of ωB97X-V is evaluated as a bilinear form over a real-space integration grid, and the cost is quadratic in the number of retained grid points. We built a scheme that keeps the exact value for the block pairs it evaluates and carries a **certified** upper bound for the ones it drops, accumulated into a single pre-declared `cert_err`, while preserving the grid-point-pair structure that a fragment-pair energy decomposition needs (P1's quantity). Then we measured it, and the measurement is the paper:

- at the accuracy the decomposition requires (1e-6 kcal/mol on a fragment-pair term), the screened path still evaluates **84.4-95.4%** of the far-field point pairs; - with no other runnable user task at the sampling instant (external R-state count 0; `load1` is recorded, 0.99-4.24, but the core count is not, so this is a contention gate and **not** a claim that the box was idle), across 24 timed runs in the archived window and 20 in a second independent one — 33 screened/exact pairs in total — the screened path is **1.08-10.28× slower** in the first window and **5.46-13.29**× in the second, never faster (§6); - the reason is visible in the bound itself: at atom-block granularity the *smallest* block-pair bound in a real water dimer exceeds the entire quantity it is meant to judge by a factor of **2.4e5**, and the near field accounts for **100%** of the cross point-pairs — every root box separation is exactly 0.0000 Å (§3, §5).

What is not new. Making this double sum cheaper is a live target. VV10 itself was introduced as a closed-form kernel replacing the adiabatic-connection integrand [1], with a separable/efficient form soon after [2]; [3], and real-space grid implementations with parallel post-processing exist [4]. The nearest current work is Laikov's preprint "Nonlocal van der Waals density functional made faster" [5] (v2, 2026-01-19), which *approximates the six-dimensional integrand by products of three-dimensional density-like distributions and potential-like functions of the interelectronic distance*, explicitly to open it to fast-multipole evaluation, with a Gaussian-fitted analytic variant on top; the claimed accuracy is empirical — its own noble-gas-dimer table gives a well-depth deviation of 103.6 μEh for XeXe, and its text assesses the tuned variant as "below 18 %" for bond energies (mostly below 1.2 % for bond lengths) — no tolerance parameter and no bound are given, and the pairwise accumulation structure is not what the method returns. Read end to end, it is a *kernel replacement*: an empirically tuned separable rational expansion standing in for VV10's Φ, which is the right move when only the total is wanted. We screen the original sum, which is the only move

available when the answer must be a pair-by-pair attribution — and we measure what preserving it costs. Efficient real-space evaluation of a nonlocal term is also established for periodic codes [6] and in the vdW-DF grid implementations cited above. Hierarchical box/MAC machinery with provable geometric error bounds is classical for Coulomb integrals and exact exchange [7]; [8]. The abbreviation "JCP" is deliberately spelled out in both halves of that pair: the two papers sit in different journals whose initials collide, and a reader who goes looking in one finds nothing. The closest *mechanism* neighbour is not a nonlocal-term paper at all: the "Linear Scaling Hierarchical Integration Scheme for the Exchange-Correlation Term" of Burow & Sierka [9] octrees the grid and screens negligible contributions adaptively — for a *single* integral over a semi-local functional, with no rigorous bound and no pair attribution. The "have a proof" half of our conjunction likewise has ancestry outside this object: rigorous multipole upper bounds for electron-repulsion integrals [10]; [11]; [12] and a rigorous relative-error criterion for tree/FMM summation of Newton potentials [13]. The object is what is unclaimed: $\iint \rho\ \Phi\ \rho$ with VV10's Φ, screened by a proven bound, with the pair ledger intact.

Publishing a refutation is not new either, and we say so deliberately: the "which fast scheme is actually faster at molecular sizes" comparison is a mature genre in electrostatics and N-body work [14]; [15]; [16]; [17]; [18], where the crossover size is known to depend on the kernel and the constants rather than on the asymptotic exponent. Our result is that same shape of measurement for a different bilinear form, and we cite these as methodological neighbours rather than as prior art on VV10.

Scope, and how confident the novelty statement is. We claim: (i) a certified per-block-pair upper bound for the VV10 real-space double sum, (ii) that it is implemented without destroying the pair attribution, and (iii) the quantitative cost measurement of (i)+(ii) against the dense kernel. We do not claim the first hierarchical or fast-multipole treatment of a nonlocal correlation term. The literature screen behind those sentences is `RESEARCH-20260930-p2-novelty.md`: 44 Web-of-Science Core Collection search strings executed through the site's own record endpoint with six known-item calibrations passing, plus Crossref, OpenAlex and arXiv, every string and hit count logged. Its limits are stated in the same file and are not small: three of the four citation-hub sweeps we wanted (Sabatini, Gulans, JuNoLo) were cut off by HTTP 429 and only the Laikov hub was enumerated to completion, Semantic Scholar was never retrieved, Scopus was not searched, and the body text of the implementation papers - where an empirical cutoff would be documented - was read only where an open copy existed. Four method names carried in our own earlier notes could not be located in any of five indexes and are therefore not named anywhere above. The novelty statement should be read as "not found in that coverage", not "does not exist"; the likeliest collision is a journal version of Laikov's preprint adding an error analysis, and this section must be re-run before submission.

3 2. Setting

The nonlocal-correlation term of a VV10-type functional is a double sum over numerical integration points,

E_NL = 1/2 Σ_i Σ_j w_i ρ_i w_j ρ_j Φ(g_ij, g'_ij), $\Phi = -3$ / (2 g g' (g + g'))

with `g = w0(r_j) R_ij`^{2} `+` κ`(r_i)` under same-centre pairing, `w0 = sqrt(C|`$\nabla\rho$`|`$^4/\rho^4$ `+ (4`π`/3)`ρ) ($\sigma = |\nabla\rho|^2$, so σ^2 enters) and κ `= b(3`π`/2)(`ρ`/9`π`)^1/6` (b = 6.0, C = 0.01) [`nlcsplit/nlc.py:31-43`; the pairing convention and its cost are established in P1 §2.1]. The reported quantity adds the local part `E_loc =` $\beta\int\rho$, β `= (1/32)(3/b`^{2}`)^3/4` [`nlcsplit/nlc.py:112`].

Two properties of this sum decide everything below.

(i) There is no sign cancellation inside a block. $\rho \geq 0$, the integration weights enter through $w\rho$, and $\Phi < 0$ everywhere, so a block sum is bounded by the sum of absolute values without loss [`nlcsplit/fastpair.py:17-18`]. Using `m_P = Σ_i∈P|w_i ρ_i|` is therefore not a looseness introduced for convenience — it is the only quantity the integrands control.

(ii) The attribution structure is the product, not the sum. We keep `pairs[(A,B)]` for atom (or fragment) blocks P, Q, so `E_pair = Σ_a v(a,a) + 2 Σ_a<b v(a,b)` and the factor of two is applied exactly once, at block level [`nlcsplit/tests/test_fastpair.py::test_x2_not_applied_twice`; the half-block convention of `nlc.pair_energy` is documented at `nlcsplit/step_d_floor.py:18`]. A point-value kernel evaluation returns `dE_NLC` and nothing else; the pair structure is what a screening scheme has to preserve to be usable for the attribution question P1 asks.

Two independent truncations, reported separately. The density floor `RHO_FLOOR = 1e-12` [`nlcsplit/nlc.py:27`] already discards points before any geometry is consulted — on the water dimer at grid level 1 it keeps 17,227 of 19,936 points [`nlcsplit/logs/step_h.json`, `nlcsplit/step_d_floor.py`]. Screening is a second, orthogonal truncation. Conflating them would make the certified error meaningless, so every number below is reported with the floor held fixed and the screening budget accounted separately.

4 3. A certified block-pair bound

For a block pair (P, Q) with axis-aligned bounding boxes, let `D` be the box separation (`D = 0` when the boxes overlap) and

a_P = w0_min(P)·D^2 + κ_min(P), bound(P,Q) = 0.75·m_P·m_Q / (a_P a_Q (a_P + a_Q)).

Then `|Σ_i∈P Σ_j∈Q w_i ρ_i w_j ρ_j Φ_ij| ≤ bound(P,Q)`, and for a same-level block pair the ×2 convention gives `mult = 2.0` [`nlcsplit/fastpair.py:239-250`, `_mac` at `:454`].

The bound is not tight by construction — it is tight *enough to be the thing we measure*. On real water-dimer fields the worst ratio over 390 sampled leaf pairs is `bound/|actual| = 1.181e+07` for the whole-block bound [`evidence/coretail/20260929T130257Z_coretail_bound.out`, Q2c, lines 134 and 147] — a factor above ten to the seventh power, which is the honest starting point for §4–§6.

What a global minimum does not buy. A bound built from one global `w0_min`/κ_min is useless rather than merely loose: at `rho_floor = 1e-12` the global values are κ_min = 0.162 and `w0_min = 2.05e-6`, so the R^{-6} asymptotic regime is only reached at 281 bohr ≈ 149 Å (one line by hand: `sqrt(kappa_min/w0_min) = sqrt(0.162/2.05e-6) = 281.1` bohr, and 281.1 × 0.529177 = 148.7 Å; the two constants come from `evidence/coretail/20260929T130257Z_coretail_bound.out`). Per-block minima are therefore mandatory, and what does the pruning is the density mass `m_P`, not the distance.

`screen` is an acceptance constant, not a budget. τ is tested per node pair, so the accumulated guarantee is `cert_err = Σ accepted bounds`, and it is `cert_err` — not τ — that must be compared with an accuracy requirement. On the benzene sandwich at level 0, taking every node pair whose individual bound is below 1e-8 Ha accepts 118,250 of them and the accumulated bound is 4.443e-4 Ha = 0.279 kcal/mol, four orders of magnitude above the per-pair threshold one was checking against [`evidence/bound_forms/20260929T101447Z_probe_bound_forms.out`, benzene block, `tau=1e-08` row]. A test pins the budget reading: requiring `cert_err ≤ 1e-9 Ha` selects τ = 1e-13

and reproduces the inter-fragment term to 2.763e-8 kcal/mol (−0.523653418 against −0.523653446) [`nlcsplit/tests/test_fastpair.py::test_L2_fragment_term_survives_screening`].

5 4. Which geometric criterion can screen at all

A top-down accept/reject is sound only if refining a block never *increases* its bound: accepting a parent pair must over-estimate the total of everything it replaces. This is a property of the geometric representation, and it is testable, so we tested it.

representation	separation used	monotonicity violations / 300,000 random refinements
mass-weighted ball (centroid + max radius)	‘D = max(0,	c_P − c_Q
axis-aligned box	per-axis gap, Euclidean norm	**0**

[`nlcsplit/tests/test_fastpair.py::test_box_separation_is_never_increased_by_refinement`, `...::test_ball_bound_can_lose_monotonicity_under_refinement`; counts recorded in `PROGRESS-nlcsplit.md` §"geometry change rationale"]

The ball failure is not numerical noise; it has a four-point hand-computable counterexample (two equal-mass point sets whose parent balls are 7.929 apart while a child pair is 5.0 apart, because a child ball protrudes outside its parent) and it is pinned as a test. The mechanism is that a mass-weighted centroid with a maximum-radius bound is not the minimum enclosing ball, so refinement can enlarge the covered region.

This is a negative result at the level of the criterion, independent of cost. A ball criterion cannot be used for top-down screening at any tolerance and on any hardware, and we initially believed the opposite: the claim that balls are "obviously" monotone under refinement was made in prose before it was counted, and the count refuted it. We state it because it is the methodological point of this section: *the choice of geometric representation for a certified scheme must be settled by a refinement-monotonicity count, not by an inclusion intuition.*

6 5. Where the tightness actually comes from

Box and ball agree exactly at the top of the hierarchy — for the water dimer the root-level bound sums are bit-identical (21723.206496881914 Ha both ways) — and at the root nothing is usable: for the benzene sandwich at level 0, **0 of 276 atom pairs** have a root box bound below 1e-8 Ha, against an exact inter-fragment total of 38.3510 kcal/mol [`evidence/bound_forms/20260929T101447Z_probe_bound_forms.out`].

The cause is geometric and it is the same in every system we ran: low-density tail points inflate each atom's bounding box until neighbouring boxes overlap, so `D = 0` at the root and the distance term contributes nothing. On the water dimer all 15 cross-atom root pairs have a box gap of exactly 0.0000 Å, and for benzene 276/276 [`evidence/bound_forms/20260929T112656Z_hybrid_nearfield.out`, "# 276/276 cross atom pairs print root box gap 0.00 A"]. That log also carries wall-clock rows; every one of them is marked `valid=False` by our own load gate (they were taken with ~10 external cores busy), so nothing in §6 comes from it — only the geometric counts. Tightness therefore arrives only by descending into blocks small enough to be simultaneously dense and compact.

We tested the two ways of getting there.

Refine below the atom block. Splitting each atom's retained points by a longest-axis bisection (stop when mass $\leq 1\%$ of the block *and* radius $\leq 1/3$ of the reference, or at 32 points) gives 1,388 nodes and 697 leaves for the water dimer at level 1, and this is the only mechanism that fires: at τ = 1e-8 Ha the tree accepts 32,131 of 197,686 leaf pairs and drops 59.7% of the far point pairs with `cert_err` = 1.278e-4 Ha $\geq$ the actual 2.346e-5 Ha [`evidence/fastpair_ab/fastpair_ab_compare.json`, `ours_tau1em8` rows].

Tighten the bound at fixed granularity. Splitting each node into a dense core and a sparse tail and bounding the three mass products separately (`m` = $\widetilde{m}$ `+ T`, core-core with core minima, mixed and tail-tail with the whole-block minima) tightens the *root* bounds by 6.1e3× (2.172e+04 → 3.552 Ha over the 15 root pairs) but the *leaf-pair* sums by only 2.13× (4.507e-2 → 2.115e-2 Ha), and the number of leaf pairs whose bound clears 1e-12 Ha by 8.8% (59,644 → 64,883) [`evidence/coretail/20260929T130257Z_coretail_bound.out`, Q2a/Q2b]. Two cautions from the same run: the variant that bounds every tail product by a distance-free global `sup|Φ|` — the form one of us specified when delegating this experiment — is **looser than the old bound** on 125,097 of 197,686 leaf pairs; and the deliberately unsound variant (tail terms using the core $\widetilde{\kappa}$) produced zero violations in 300k+ random samples and had to be refuted with a two-point hand fixture, where it underestimates by 42×. A negative control with no power is not a test.

7 6. What the screening costs with no other runnable task

Sections 3-5 say the bound is loose and that tightening it at fixed granularity buys 1.11×. Neither of those is a statement about time, so we measured time.

Protocol. Water chains of 1-4 monomers (3, 6, 9 and 12 atoms), ωB97X-V/6-31G (cartesian), PySCF 2.12.1 [19], `OMP_NUM_THREADS=4`. For each system we ran the exact double sum and the same code with screening at τ = 1e-8, 1e-10 and 1e-12 Ha, each after its own converged SCF, on the same grid. The one thing that made this measurable was a load gate: a sample is only valid if the number of *Runnable* tasks outside our own process tree is ≤ 1 at the moment the timed block ends. Two earlier attempts produced numbers we threw away — one headline ("5.2× slower") was taken while twelve agents were contending on the box, and before that the gate counted our own work as contention, which made every sample invalidate itself. All 24 samples in [`evidence/scaling/step_j.json`] carry `valid_timing: true`.

Result: the screened path never won, anywhere in the ladder. Ratios are screened/exact wall clock, quoted with the fraction of far point pairs the screened path still evaluated, and the absolute error it made on the pair term:

system (level 0)	**exact (s)**	**τ=1e-8**	**τ=1e-10**	**τ=1e-12**
3 atoms	0.08	5.31×, 72.6%, 1.3e-3	6.89×, 88.9%, 1.2e-5	7.98×, 95.4%, 6.8e-8
6 atoms	0.48	2.93×, 55.1%, 7.1e-3	4.95×, 81.9%, 4.6e-5	5.42×, 93.7%, 1.2e-7
9 atoms	0.52	5.66×, 46.1%, 1.4e-2	8.89×, 74.5%, 1.4e-4	10.25×, 91.3%, 6.1e-7
12 atoms	0.99	4.65×, 37.5%, 2.4e-2	8.42×, 66.2%, 3.0e-4	10.28×, 86.3%, 1.8e-6
system (level 1)	exact (s)			
6 atoms	4.45	1.09×, 41.6%, 1.5e-2	1.64×, 68.4%, 7.9e-5	1.95×, 84.4%, 4.4e-7
9 atoms	9.63	1.08×, 31.4%, 3.8e-2	1.63×, 60.5%, 3.1e-4	2.19×, 79.5%, 1.5e-6

Two readings matter more than the headline ratio.

τ is not a budget. P1 needs each pair term right to 1e-6 kcal/mol. At τ = 1e-12 the 12-atom level-0 chain is off by 1.8e-6 and the 9-atom level-1 chain by 1.5e-6 — both outside that, so no single

τ in this sweep is admissible across the ladder. The reason is visible in `cert_err`, the accumulated guarantee, which grows with system size at fixed τ: 2.8e-10 $\rightarrow$ 1.2e-9 $\rightarrow$ 3.4e-9 $\rightarrow$ 9.2e-9 Ha for 3 $\rightarrow$ 12 atoms at level 0, and 1.4e-8 Ha for 9 atoms at level 1, because it is a sum over accepted node pairs and a bigger chain has more of them. A per-pair acceptance constant therefore cannot promise a per-quantity error; only the accumulated bound can, and it is the thing that has to be driven below target — which pushes τ toward 1e-13, where 95+% of the pairs are evaluated anyway.

The traversal is the cost, and it is paid to keep almost all of the work. The tree counters for the two cheapest-ratio cases (level 1) show the arithmetic plainly: the 9-atom chain visits 391,955 node pairs to decide about 493,395 leaf pairs, and then evaluates 348,536 of those leaf pairs — 70.6% of them — which is 236,127,947 of 297,165,609 far point pairs, 79.5% of the original work, plus 391,955 bound evaluations and a 1,061-node tree built from scratch. Skipping 20% of a gemm by doing 400k scalar bound computations in Python-adjacent numpy is a losing trade at these sizes, and the loss is largest exactly where the ratio looked best: at level 1 τ = 1e-8 the screened path is 1.08$\times$ slower and wrong by 3.8e-2 kcal/mol.

We deliberately do not plot a scaling curve. An asymptotic claim here would need a protocol we did not run: τ is an absolute per-node-pair acceptance constant, and we measured that the accumulated guarantee `cert_err` grows with system size at fixed τ (2.8e-10 $\rightarrow$ 9.2e-9 Ha from 3 to 12 atoms at level 0), so this definition of the scheme has no clean scaling limit — to stay inside a fixed error the budget per pair must shrink as pairs accumulate, and because the quantity being bounded decays only as a power of the inter-block distance, the certified cutoff can then grow only like a fractional power of `N`, which puts the pair count back near quadratic. We did not measure that exponent and we are not claiming it. What we claim is the finite-size measurement: at the accuracy this decomposition is used for, on chains up to 12 atoms and grids up to 30,000 points, every one of the 33 timed screened/exact pairs is slower, by 1.08-10.28$\times$ in the archived window and 5.46-13.29$\times$ in the second. The figure ([`figs/fig3_screening_cost.png`]) therefore plots wall clock against far point pairs *actually evaluated*, per system, which makes the statement "for the same amount of work the screened path takes longer" checkable without any appeal to a limit.

8 7. What would have to be true for this to pay

Stated as conditions we did not meet, so the reader can check them against their own system before implementing anything:

1. **The bound must clear the target at coarse granularity.** At atom-pair granularity the *smallest* block-pair bound we found in a real dimer is 197.96 Ha against a quantity of $-$8.34e-4 Ha (`n_pairs_bound_below_value = 0`, `dropped_pairs = []`) [`evidence/fastpair_ab/fastpair_ide_validation.json`]. No traversal can fix a bound that is loose by 2.4e5$\times$; it can only descend, and descent is what costs. 2. **Most pairs must be droppable at the target accuracy.** At the coarsest τ in the sweep that meets `|dE|` $\leq$ `1e-6` kcal/mol, the screened path still evaluates 84.4-95.4% of the far point pairs (droppable share 4.6-15.6%) — and for two of the six systems (12 atoms at level 0, 9 atoms at level 1) no τ in the sweep meets the target at all, so their admissible τ is finer and the share smaller. 3. **The per-pair decision must be cheaper than the arithmetic it avoids.** Measured false: the tree is cheaper than the large gemm on **0 of 15** root pairs, and the near field is 100% of the cross point pairs (all 15 root box gaps are exactly 0.0000 Å). 4. **The system must have a far field at all.** Our ladder is a chain at fixed density; compact dimers (S22) have no inter-block region

that is both distant and negligible. A 300-atom protein-ligand complex with genuinely separated fragments is outside what we measured, and we make no claim about it.

Condition 1 is structural for VV10 as implemented in PySCF: the kernel's `g` is `w0(r_j)` R^2 + κ`(r_i)`, and `w0` contains `(4`π`/3)`ρ under a square root, so a *low-density* grid point carries a small `w0` and a huge bound. The tail points that inflate the boxes are the same points that fail to make the bound decay. That coupling is why the atom-block bound is loose, why refinement is the only mechanism that fires, and why refinement does not pay.

References

[1] Oleg A. Vydrov and Troy Van Voorhis. Nonlocal van der Waals density functional made simple. *Physical Review Letters*, 103:063004, 2009. doi: 10.1103/physrevlett.103.063004.

[2] Oleg A. Vydrov and Troy Van Voorhis. Nonlocal van der Waals density functional: The simpler the better. *The Journal of Chemical Physics*, 133:244103, 2010. doi: 10.1063/1.3521275.

[3] Riccardo Sabatini, Tommaso Gorni, and Stefano de Gironcoli. Nonlocal van der Waals density functional made simple and efficient. *Physical Review B*, 87:041108, 2013. doi: 10.1103/physrevb.87.041108.

[4] Predrag Lazić, Nicolae Atodiresei, Mojtaba Alaei, Vasile Caciuc, Stefan Blügel, and Radovan Brako. JuNoLo – Jülich nonlocal code for parallel post-processing evaluation of vdW-DF correlation energy. *Computer Physics Communications*, 181:371–379, 2010. doi: 10.1016/j.cpc.2009.09.016.

[5] Dimitri N. Laikov. Nonlocal van der Waals density functional made faster, 2026. URL `http://arxiv.org/abs/2511.04169v2`. v2, updated 2026-01-19 (v1 submitted 2025-11-06); preprint, no DOI in Crossref.

[6] Dmitrii Nabok, Peter Puschnig, and Claudia Ambrosch-Draxl. noloco: An efficient implementation of van der Waals density functionals based on a Monte-Carlo integration technique. *Computer Physics Communications*, 182:1657–1662, 2011. doi: 10.1016/j.cpc.2011.04.015.

[7] Eric Schwegler, Matt Challacombe, and Martin Head-Gordon. A multipole acceptability criterion for electronic structure theory. *The Journal of Chemical Physics*, 109:8764–8769, 1998. doi: 10.1063/1.477546.

[8] Walter Dehnen. A Hierarchical (N) Force Calculation Algorithm. *Journal of Computational Physics*, 179:27–42, 2002. doi: 10.1006/jcph.2002.7026.

[9] Asbjörn M. Burow and Marek Sierka. Linear scaling hierarchical integration scheme for the exchange-correlation term in molecular and periodic systems. *Journal of Chemical Theory and Computation*, 7:3097–3104, 2011. doi: 10.1021/ct200412r.

[10] Daniel S. Lambrecht and Christian Ochsenfeld. Multipole-based integral estimates for the rigorous description of distance dependence in two-electron integrals. *The Journal of Chemical Physics*, 123:184101, 2005. doi: 10.1063/1.2079967.

[11] Daniel S. Lambrecht, Bernd Doser, and Christian Ochsenfeld. Rigorous integral screening for electron correlation methods. *The Journal of Chemical Physics*, 123:184102, 2005. doi: 10.1063/1.2079987.

[12] Bernd Doser, Daniel S. Lambrecht, and Christian Ochsenfeld. Tighter multipole-based integral estimates and parallel implementation of linear-scaling AO–MP2 theory. *Physical Chemistry Chemical Physics*, 10:3335, 2008. doi: 10.1039/b804110e.

[13] Stefan Engblom. On well-separated sets and fast multipole methods. *Applied Numerical Mathematics*, 61:1096–1102, 2011. doi: 10.1016/j.apnum.2011.06.011.

[14] Brock A. Luty, Malcolm E. Davis, Ilario G. Tironi, and Wilfred F. Van Gunsteren. A comparison of particle-particle, particle-mesh and Ewald methods for calculating electrostatic interactions in periodic molecular systems. *Molecular Simulation*, 14:11–20, 1994. doi: 10.1080/08927029408022004.

[15] Henrik G. Petersen. Accuracy and efficiency of the particle mesh Ewald method. *The Journal of Chemical Physics*, 103:3668–3679, 1995. doi: 10.1063/1.470043.

[16] Markus Deserno and Christian Holm. How to mesh up Ewald sums. I. a theoretical and numerical comparison of various particle mesh routines. *The Journal of Chemical Physics*, 109:7678–7693, 1998. doi: 10.1063/1.477414.

[17] Nail A. Gumerov and Ramani Duraiswami. Fast multipole methods on graphics processors. *Journal of Computational Physics*, 227:8290–8313, 2008. doi: 10.1016/j.jcp.2008.05.023.

[18] Bartosz Kohnke, Carsten Kutzner, and Helmut Grubmüller. A GPU-Accelerated fast multipole method for GROMACS: Performance and accuracy. *Journal of Chemical Theory and Computation*, 16:6938–6949, 2020. doi: 10.1021/acs.jctc.0c00744.

[19] Qiming Sun, Xing Zhang, Samragni Banerjee, Peng Bao, Marc Barbry, Nick S. Blunt, Nikolay A. Bogdanov, George H. Booth, Jia Chen, Zhi-Hao Cui, Janus J. Eriksen, Yang Gao, Sheng Guo, Jan Hermann, Matthew R. Hermes, Kevin Koh, Peter Koval, Susi Lehtola, Zhendong Li, Junzi Liu, Narbe Mardirossian, James D. McClain, Mario Motta, Bastien Mussard, Hung Q. Pham, Artem Pulkin, Wirawan Purwanto, Paul J. Robinson, Enrico Ronca, Elvira R. Sayfutyarova, Maximilian Scheurer, Henry F. Schurkus, James E. T. Smith, Chong Sun, Shi-Ning Sun, Shiv Upadhyay, Lucas K. Wagner, Xiao Wang, Alec White, James Daniel Whitfield, Mark J. Williamson, Sebastian Wouters, Jun Yang, Jason M. Yu, Tianyu Zhu, Timothy C. Berkelbach, Sandeep Sharma, Alexander Yu. Sokolov, and Garnet Kin-Lic Chan. Recent developments in the PySCF program package. *The Journal of Chemical Physics*, 153: 024109, 2020. doi: 10.1063/5.0006074.